

Magic Flute

Our project is a pseudo wind instrument resembling something like a flute or recorder. It uses 3D printed “keys” (See photos section) to capture what notes to play, and uses a mic to determine whether the user is blowing to simulate blowing into the flute. The flute is displayed on the VGA display with its corresponding holes covered to match what is happening in real time. There is also a record feature that allows users to record and play back a sequence of notes.

What makes this a magic flute is that it can embody many different instruments. There is the regular flute, a clarinet, an oboe, an animal flute, and a fah flute. If you begin hyperventilating from blowing into the mic too hard, there is also a switch that magically curses you with infinite breath.

Operation

1. Press any key to begin. Pressing KEY1 at any point will bring you back to the start screen.
2. Select either blowing mode or infinite breath mode by toggling SW9.
3. If SW9 is high, play the flute to your heart's desire.
4. If SW9 is low, blow into the microphone to play the flute.
5. Switch instruments by setting SW[2:0] to 0b001, 0b010, 0b011, 0b100, or 0b101 and then pressing KEY0.
6. Toggle SW3 to 1 or high and press KEY2 to begin recording what you are playing.
7. Press KEY3 to play back what you have recorded.
8. Once the playback is done, you can go back to playing like normal.
9. For an example of the operation, you can view the demo video in the Photos section.

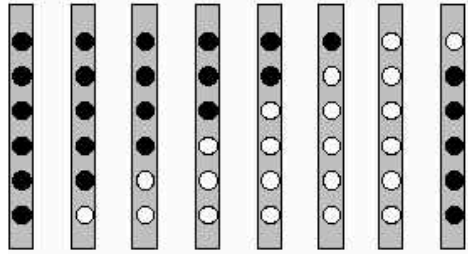
<u>Some songs you can play</u>	<u>Fingering chart for valid notes</u>
● Chariots of Fire ● Ode to Joy ● Do - re - mi ● Joy to the World ● Hot Cross Buns ● My Heart Will Go On intro ● Animal sounds ● Fah sound effect	 (inspired by the Bansuri instrument)

Figure 1. Song Suggestion and Fingering Chart

Block Diagram

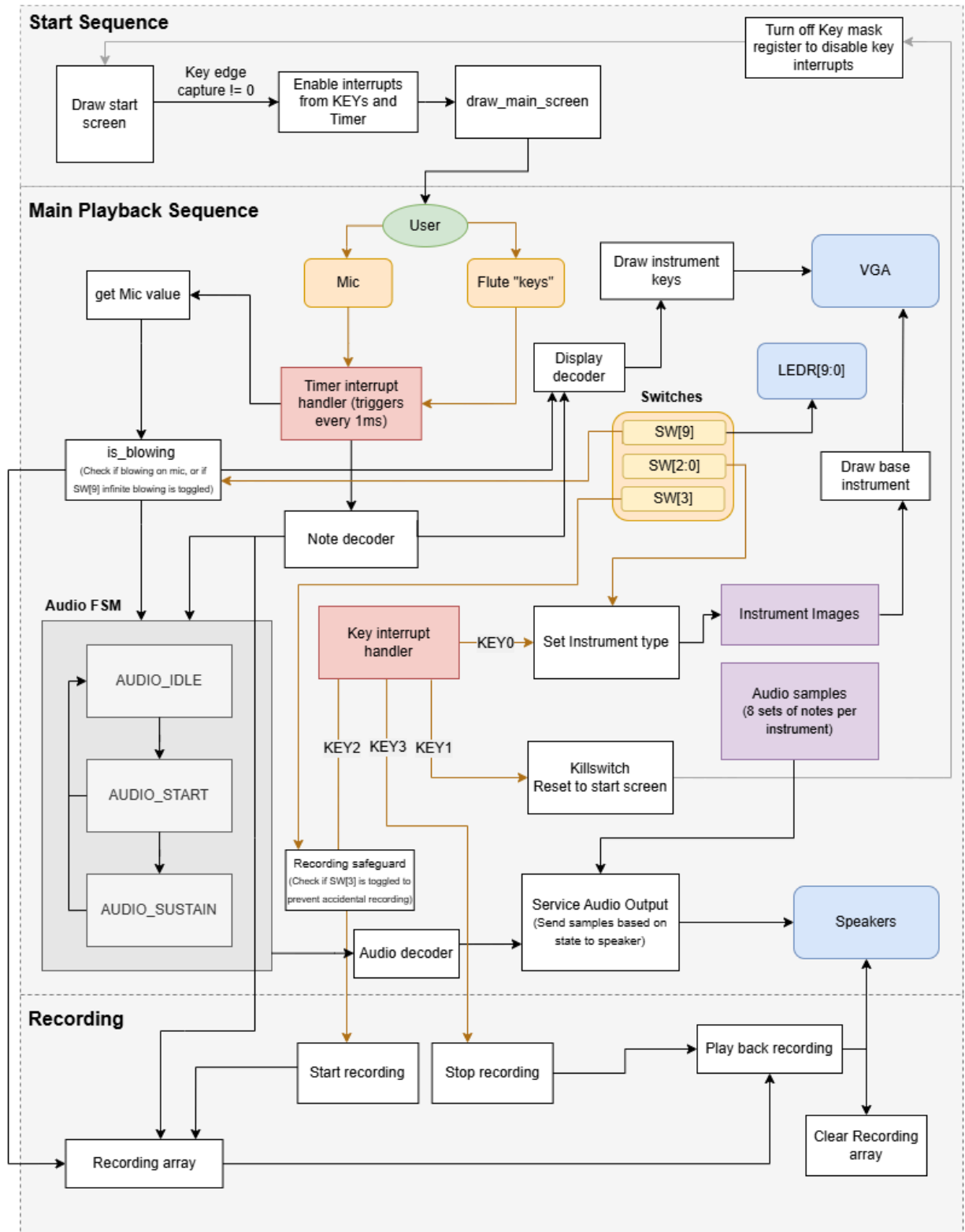


Figure 2. Block Diagram

Photo and video

Video:

https://drive.google.com/file/d/1_hJxVGFBxOSFgxUjt5JGT5bazFpn7YT/view?usp=sharing

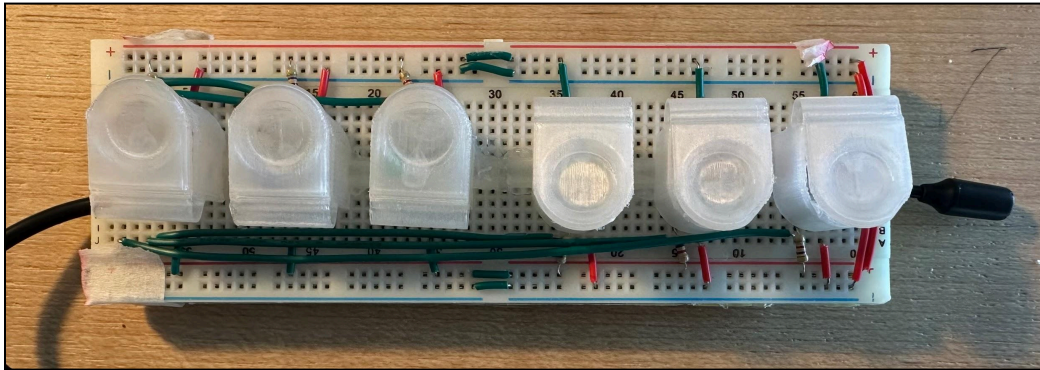


Figure 3. Flute

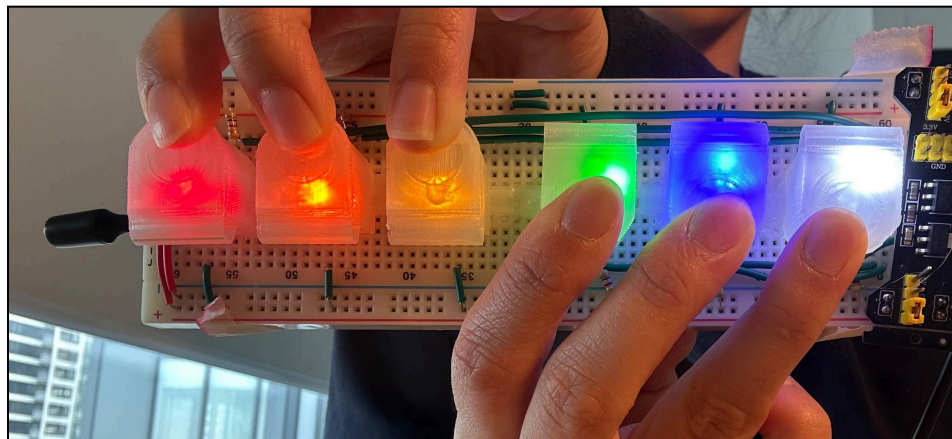


Figure 4. Flute lit up

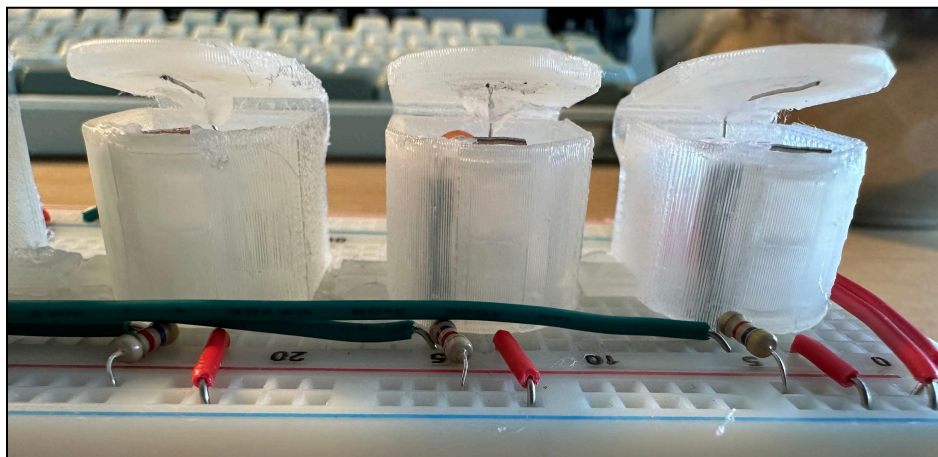


Figure 5. Close up of flute key

Attribution Table

Tanatswa	Alyssa
Drawing covered instruments, labels and normal instruments, found the images for the flute and clarinet displays	Drawing of start screen animation, creating the pixel images for animal flute, oboe, fah flute and background borders
Audio recording feature	Audio FSM for normal playback
Getting audio samples of clarinet, oboe, animal	Getting audio samples of flute and fah
Integrating audio recording, audio FSM, interrupts and the image drawing	Building hardware flute using LEDs and 3D modeling + printing keys
Audio and display decoders	Connecting microphone and GPIO pins to Audio FSM
Interrupts for Keys	Interrupts for 1ms interval timer

Figure 6. Attribution Table

References and Accreditations

Clarinet, Oboe samples by MTG - <https://freesound.org/people/MTG/>
Cat Meow, Wolf Howling, Goat Sound, Cow Moo by Dragon Studio - https://pixabay.com/users/dragon-studio-38165424/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=515977
Chipmunk Chatter by Freesound Community - https://pixabay.com/users/freesound_community-46691455/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=99627
Hyena Laugh from Sound Dino - <https://sounddino.com/en/effects/hyena/>
Flute samples by John Clements <https://www.brinckerhoff.org/flute-tones/>
Clarinet Icon by Freepik - <https://www.flaticon.com/authors/freepik>
Flute Icon by Umeicon - <https://www.flaticon.com/authors/umeicon>